

On the Influence of Delay Spread in TDL and CDL Channel Models for Downlink 5G MIMO Systems

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Abstract—5G is the next mobile generation expected to be commercially available by 2020. This work addresses the influence of the delay spread in TDL and CDL channel models for a downlink 5G MIMO system, in order to better understand their behavior. The performance is evaluated in terms of BER and throughput through link-level simulations. Results show how the different DS values impact the performance of the system when using TDL and CDL channels and are valuable when designing 5G communication systems.

Index Terms—Mobile communication systems, 5G, delay spread, TDL, CDL, throughput, BER

I. INTRODUCTION

The last decades have shown a continuous evolution of mobile communication systems. The number of users, devices, data traffic and new use cases have experienced an explosive growth over the last years. The Fifth Generation (5G) is the next generation of mobile communication systems and it is expected to revolutionize our society, individual lives and economy. Among the specified requirements proposed for 5G are extremely high data rates, reaching up to 20 Gbps, increased energy and spectrum efficiency, mobility up to 500 km/h and reduced latency under 1 ms or lower [1]. Additionally, when compared to the current generation that was primarily designed to offer high speed mobile broadband, 5G networks will be able to support new use-cases and applications. The three major use-cases established for 5G are [2]:

- Enhanced mobile broadband (eMBB): bringing an improvement on the end-user experience and performance, accompanied with a massive growth of mobile data.
- Massive machine-type communication (mMTC): representing the connectivity between billions of objects, sensors and machines. These objects are expected to be relatively low cost with a long battery life.
- Ultra-reliable and low latency communication (URLLC): requiring an infrastructure that offers ultra-reliability, availability and low latency intended primarily for mission critical services.

In the future, more use-cases will emerge that will require flexible 5G networks with the ability to adapt to new requirements and expectations.

The Delay Spread (DS) constitutes an important factor to take into consideration when using frequency selective channels since it is one of the most important parameters that characterizes a frequency selective fading channel. It is extracted from the Power Delay Profile (PDP). Specifically, it corresponds to the second central moment of the PDP and it measures the time difference from the arrival of the initial multi-path component, which is generally the Line-of-Sight (LOS) component, and the arrival of the last multi-path component [3].

The main objective of this paper is to evaluate the influence of the DS on the performance of physical layer using Tapped Delay Line (TDL) and Clustered Delay Line (CDL) channel models and a modulation scheme of 256 Quadrature Amplitude Modulation (QAM) for the future 5G systems. The significant contributions of this paper include the evaluation of 5G communication system's performance in terms of Bit Error Rate (BER) and user throughput. Moreover, simulation results show how the different channels affect the system performance in different environments with specific DSs selected.

The remainder of this paper is organized as follows, Section II presents the TDL and CDL channel models, Section III describes the simulation scenarios investigated and Section IV presents the simulation results and analysis. Lastly, Section V provides the conclusions of this paper.

II. TDL AND CDL CHANNEL MODELS

The 3rd Generation Partnership Project (3GPP) released a technical report on channel models for frequencies from 0.5 to 100 GHz envisioned for the future 5G communication systems. The purpose of the document is to assist in modeling and evaluation of the performance of physical layer techniques, using suitable channel models. There are two channel models proposed for link-level simulations for 5G for the full frequency range from 0.5 to 100 GHz: TDL and CDL models, which are addressed individually below.

A. TDL Channel Model

The TDL channel model has 5 different profiles, depending on the environment scenario desired. For Non-Line-of-Sight (NLOS) scenarios, the existing profiles are TDL_A, TDL_B and TDL_C. For LOS, the associated profiles are TDL_D and TDL_E. The channel impulse response of the TDL channel model can be defined as [5]:

$$H(t, \tau) = \sum_{i=1}^n a_i(t) \delta(\tau - \tau_i) \quad (1)$$

where $a_i(t)$ refers to the amplitude at the τ_i delay for the i th tap. The NLOS designed profiles are defined by having 23 taps, with exception for TDL_C, which has 24 taps. The LOS profiles, TDL_D and TDL_E, have 13 and 14 taps, respectively. The normalized delay and relative power for each tap varies from one profile to another and the specifications of each profile can be found in [4].

B. CDL Channel Model

The CDL channel models are an extension of the TDL profiles, and are designed for 3D channels. Therefore, there are also five different CDL profiles. For NLOS, there are three CDL models, specifically CDL_A, CDL_B and CDL_C. For LOS environments, the associated CDL models are CDL_D and CDL_E.

When generating the channel coefficients for the CDL profiles, the first step is to generate the azimuth angles of departure and arrival. Hence, the arrival angles for the m th ray of the n th cluster can be obtained by [4]:

$$\phi_{n,m,AoA} = \phi_{n,AoA} + c_{ASA} \alpha_m, \quad (2)$$

where $\phi_{n,AoA}$ represents the Azimuth Angle of Arrival (AoA), c_{ASA} is the azimuth spread of arrival angle and α_m defines the ray offset angle within a cluster. To generate the angles of departure, a similar equation is used [4]:

$$\phi_{n,m,AoD} = \phi_{n,AoD} + c_{ASD} \alpha_m, \quad (3)$$

where $\phi_{n,AoD}$ represents the Azimuth Angle of Departure (AoD) and c_{ASD} is the cluster-wise Root Mean Square (RMS) azimuth spread of departure angle. In addition, the generation of the $\theta_{n,m,ZoA}$ Zenith Angle of Arrival (ZoA) and $\theta_{n,m,ZoD}$ Zenith Angle of Departure (ZoD) angles is identical to the equations (1) and (2), respectively. The $\phi_{n,m,AoD}$ AoD angles are randomly coupled with the $\phi_{n,m,AoA}$ AoA angles within a cluster. The $\theta_{n,m,ZoD}$ ZoD angles are coupled with the $\theta_{n,m,ZoA}$ ZoA angles and the $\phi_{n,m,AoD}$ AoD angles are also randomly coupled with the $\theta_{n,m,ZoD}$ ZoD angles. Following, the Cross Polarization Power Ratio (XPR), k , is derived for each ray m of each cluster n :

$$k = 10^{C/10}, \quad (4)$$

where C is the XPR for each cluster in dB. Finally, the channel coefficients for each cluster n and pair of transmitter and

receiver antenna elements (t, r) for a NLOS environment are given by [6]:

$$H_{r,t,n}^{NLOS}(t) = \sqrt{\frac{P_n}{M}} \sum_{m=1}^M \begin{bmatrix} F_{rx,r,\theta}(\theta_{n,m,ZoA}, \phi_{n,m,AoA}) \\ F_{rx,r,\phi}(\theta_{n,m,ZoA}, \phi_{n,m,AoA}) \end{bmatrix}^T \times \begin{bmatrix} e^{j\Phi_{n,m}^{\theta\theta}} \sqrt{k_{n,m}^{-1}} e^{j\Phi_{n,m}^{\theta\phi}} \\ \sqrt{k_{n,m}^{-1}} e^{j\Phi_{n,m}^{\phi\theta}} e^{j\Phi_{n,m}^{\phi\phi}} \end{bmatrix} \begin{bmatrix} F_{tx,t,\theta}(\theta_{n,m,ZoD}, \phi_{n,m,AoD}) \\ F_{tx,t,\phi}(\theta_{n,m,ZoD}, \phi_{n,m,AoD}) \end{bmatrix} \times e^{\frac{j2\pi(\hat{r}_{rx,n,m}^T \cdot \bar{d}_{rx,r})}{\lambda_0}} \times e^{\frac{j2\pi(\hat{r}_{tx,n,m}^T \cdot \bar{d}_{tx,t})}{\lambda_0}} \times e^{\frac{j2\pi(\hat{r}_{rx,n,m}^T \cdot \bar{v})}{\lambda_0}} t, \quad (5)$$

where $F_{tx,t,\theta}$ and $F_{tx,t,\phi}$ represent the field radiation patterns of the transmit antenna element, t , and $F_{rx,r,\theta}$ and $F_{rx,r,\phi}$ are the field radiation patterns of the receiver antenna element, r . The last line of the above equation corresponds to the mobile terminal phase array offset, the g-NodeB phase array offset and the Doppler phase shift. From eq. 5, $\bar{d}_{rx,r}$ and $\bar{d}_{tx,t}$ define the location vector of the receiver and transmitter antenna element r and t , respectively. λ_0 is the wavelength of the carrier frequency and $\hat{r}_{rx,n,m}$ indicates the spherical unit vector with the azimuth arrival angle and zenith arrival angle, $\phi_{n,m,AoA}$ and $\theta_{n,m,ZoA}$, respectively. It is given by [7]:

$$\hat{r}_{rx,n,m} = \begin{bmatrix} \sin \theta_{n,m,ZoA} & \cos \phi_{n,m,AoA} \\ \sin \theta_{n,m,ZoA} & \sin \phi_{n,m,AoA} \\ \cos \theta_{n,m,ZoA} \end{bmatrix} \quad (6)$$

A similar expression for the spherical unit vector, $\hat{r}_{tx,n,m}$, with the azimuth and zenith departure angle, $\phi_{n,m,AoD}$ and $\theta_{n,m,ZoD}$, respectively, is given [7]:

$$\hat{r}_{tx,n,m} = \begin{bmatrix} \sin \theta_{n,m,ZoD} & \cos \phi_{n,m,AoD} \\ \sin \theta_{n,m,ZoD} & \sin \phi_{n,m,AoD} \\ \cos \theta_{n,m,ZoD} \end{bmatrix} \quad (7)$$

The channel coefficients for a LOS environment are given by [8]:

$$H_{r,t,1}^{LOS}(t) = \begin{bmatrix} F_{rx,r,\theta}(\theta_{LOS,ZoA}, \phi_{LOS,AoA}) \\ F_{rx,r,\phi}(\theta_{LOS,ZoA}, \phi_{LOS,AoA}) \end{bmatrix}^T \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} F_{tx,t,\theta}(\theta_{LOS,ZoD}, \phi_{LOS,AoD}) \\ F_{tx,t,\phi}(\theta_{LOS,ZoD}, \phi_{LOS,AoD}) \end{bmatrix} e^{\frac{j2\pi(\hat{r}_{rx,LOS}^T \cdot \bar{d}_{rx,r})}{\lambda_0}} \times e^{\frac{j2\pi(\hat{r}_{tx,LOS}^T \cdot \bar{d}_{tx,t})}{\lambda_0}} \times e^{\frac{j2\pi(\hat{r}_{rx,LOS}^T \cdot \bar{v})}{\lambda_0}} t \times e^{\frac{j2\pi d_{3D}}{\lambda_0}} \quad (8)$$

III. SIMULATION SCENARIOS

In this section we present the scenarios utilized for the link-level simulations. We consider a downlink single-user Multiple-Input Multiple-Output (MIMO) system with 1 Base Station (BS) equipped with 4 transmitter antennas and 1 Mobile Station (MS) also equipped with 4 receiver antennas, moving at a speed of 3 km/h. Minimum Mean Square Error (MMSE) estimation is used at the receiver part. A carrier frequency of 3.5 GHz is selected with 100 MHz of bandwidth.

The number of Physical Resource Blocks (PRBs) allocated is 256 and the duplex mode chosen is Time Division Duplex (TDD). The modulation scheme adopted is 256 QAM modulation. The link-level simulations were effectuated using Nokia's internal simulator.

Additionally, a delay scaling factor is introduced to both TDL and CDL channel models used for the simulations. Therefore, the RMS delay spread values are normalized and scaled in delay, and can be obtained by [5]:

$$\tau_{n,scaled} = \tau_{n,model} \cdot DS_{desired}, \quad (9)$$

where $\tau_{n,model}$ represents the normalized delay value for the n th cluster of the TDL or CDL model, $DS_{desired}$ is the desired DS and $\tau_{n,scaled}$ is the new delay value of the n th cluster. As mentioned earlier, the objective of this paper is to study the influence of the DS while using TDL and CDL channels intended for 5G communication systems. Consequently, the scenarios adopted for the link-level simulations are defined below in Fig. 1-3. To note that all scenarios use real channel estimation.

#	Environment	Scenario	Channel Model	Proposed DS [ns]
1	NLOS	Indoor office	TDL_A	18
2	NLOS	Indoor office	TDL_B	18
3	NLOS	Indoor office	TDL_C	18
4	LOS	Indoor office	TDL_D	18
5	LOS	Indoor office	TDL_E	18
6	NLOS	Indoor office	CDL_A	18
7	NLOS	Indoor office	CDL_B	18
8	NLOS	Indoor office	CDL_C	18
9	LOS	Indoor office	CDL_D	18
10	LOS	Indoor office	CDL_E	18

Fig. 1. 1st set of scenarios for the link-level simulations.

#	Environment	Scenario	Channel Model	Proposed DS [ns]
11	LOS	Indoor office	TDL_E	18
12	LOS	Indoor office	TDL_E	32
13	LOS	Indoor office	TDL_E	53
14	LOS	RMa	TDL_E	34
15	LOS	RMa	TDL_E	39

Fig. 2. 2nd set of scenarios for the link-level simulations.

#	Environment	Scenario	Channel Model	Proposed DS [ns]
16	NLOS	UMi Street-canyon	CDL_A	47
17	NLOS	UMi Street-canyon	CDL_A	95
18	NLOS	UMi Street-canyon	CDL_A	318
19	NLOS	UMa	CDL_A	365
20	NLOS	UMa	CDL_A	1150

Fig. 3. 3rd set of defined scenarios for the link-level simulations.

IV. SIMULATION RESULTS

This section provides the link-level simulation results and discussion. Primarily, all TDL and CDL channels's performance is compared, using a DS of 18 ns, simulating an indoor scenario. The performance of the different scenarios is evaluated in terms of BER and throughput. Fig.4 and Fig.5 present the simulation results of the scenarios 1-5, in terms of BER and throughput, respectively. BER results show that

for NLOS, the TDL channel profile with the best performance is TDL_A, followed by TDL_B and lastly TDL_C. For LOS, up to 34 dB, TDL_D performs slightly better than TDL_E, albeit for higher values of Signal-to-Noise-Plus-Interference Ratio (SNIR) TDL_E has an overall increased performance when compared to TDL_D.

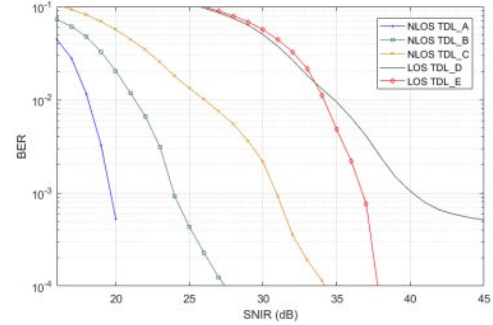


Fig. 4. BER results for all TDL profiles in an indoor scenario using a DS of 18 ns.

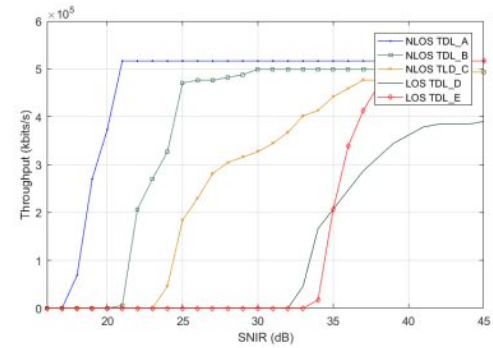


Fig. 5. Throughput results for all TDL profiles in an indoor scenario using a DS of 18 ns.

From Fig.5 we can observe that TDL_A is the first channel profile to reach maximum throughput of 516.5 Mbits/s from the NLOS profiles, meanwhile for LOS the profile TDL_E performs better than TDL_D for higher than 35 dB.

Fig.6 and Fig.7 depict the performance results for the scenarios 6-10, simulated with the same conditions as of Fig.4 and Fig.5. From both figures, we can conclude that the CDL profiles that perform better are CDL_A and CDL_E, for NLOS and LOS, respectively.

It is important to notice that for both channel models, the NLOS profiles perform better than the LOS models. This is due to the fact that the NLOS profiles have a higher number of taps/clusters (depending if we are using TDL or CDL channel model), which leads to a higher number of reflexions. Moreover, when using multi-path channels, the reconstruction of the transmitted signal at the reception is easier when the number of reflexions is higher, since the probability of having constructive interference is also higher. Consequently, in order to investigate the effect of the DS in TDL and CDL channels, we selected the best performing TDL profile for LOS environments and the best CDL profile for NLOS scenarios, TDL_E and CDL_A, respectively.

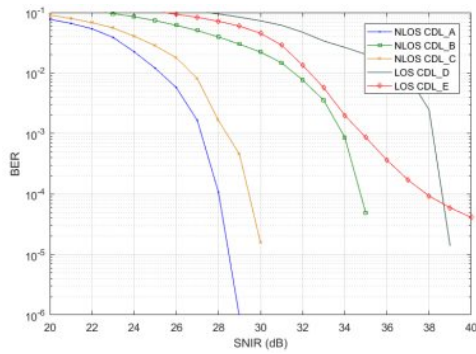


Fig. 6. BER results for all CDL profiles in an indoor scenario using a DS of 18 ns.

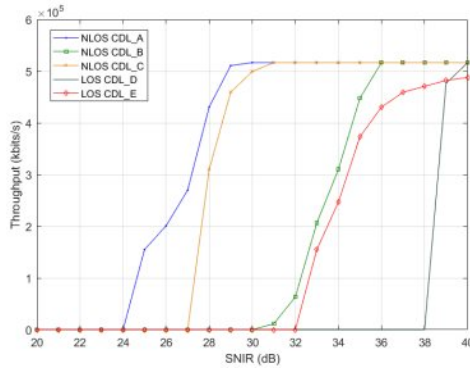


Fig. 7. Throughput results for all CDL profiles in an indoor scenario using a DS of 18 ns.

Fig.8 and Fig.9 depict BER and throughput simulation results for the TDL_E model in two different scenarios: Indoor Office (IO) and Rural Macro-cell (RMA), both with short delay spreads, ranging from 18 ns to 53 ns. Results show that above a certain threshold (in this case around 35 dB), with better conditions, the influence of the DS is not so pertinent for the values investigated. Moreover, the signal's level is much harder to be attained in a real environment, since such high values for SNIR translate to nearly ideal environment conditions. Otherwise, the behavior of the simulated system in both IO and RMA scenarios is as expected, for both BER and throughput.

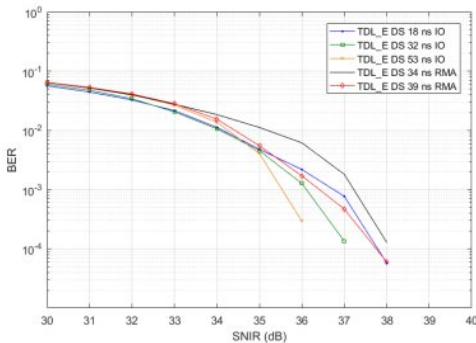


Fig. 8. BER results for TDL_E profile in an indoor office scenario using a DS of 18,32,53 ns and in a rural macrocell scenario using a DS of 34,39 ns.

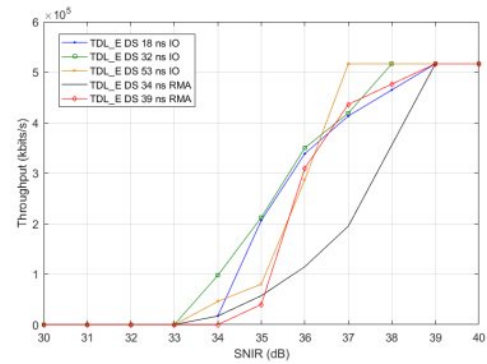


Fig. 9. Throughput results for TDL_E profile in an indoor office scenario using a DS of 18,32,53 ns and in a rural macrocell scenario using a DS of 34,39 ns.

Fig.10 and Fig.11 depict BER and throughput simulation results for the CDL_A model in two different scenarios: Urban Micro-cell Street Canyon and Urban Macro-cell (UMA), both with normal and long delay spreads, ranging from 47 ns to 1150 ns. The performance of the system is better when considering the street canyon scenario compared to the UMA, which is composed by much higher DS values. However, we can observe that after a threshold of approximately 32 dB (extremely good real environment conditions), the influence of the DS is not so relevant. Moreover, the system performs the worst when using a DS of 1150 ns, within the UMA scenario.

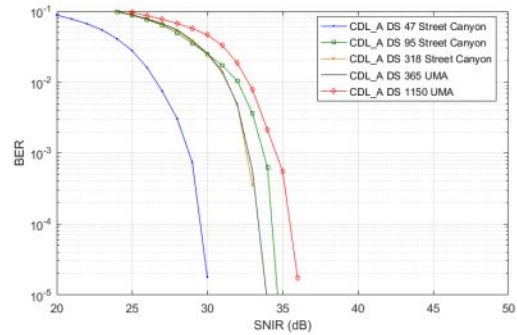


Fig. 10. BER results for CDL_A profile in a street canyon scenario using a DS of 47,95,318 ns and in an urban macrocell scenario using a DS of 365,1150 ns.

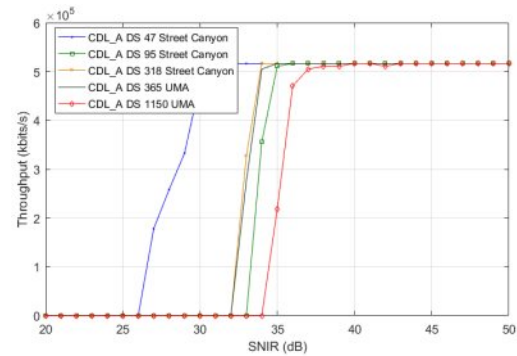


Fig. 11. Throughput results for CDL_A profile in a street canyon scenario using a DS of 47,95,318 ns and in an urban macrocell scenario using a DS of 365,1150 ns.

V. CONCLUSION

This paper addressed and evaluated the influence of the DS in the performance of a 5G communication system using TDL and CDL channel models proposed by 3GPP, in terms of BER and user throughput. Link-level simulations were performed using Nokia's internal simulator and the results obtained enabled a better understanding on how the DS influences a 5G communication system and how the different channel models studied affect its performance within the different scenarios evaluated (indoor office, rural macro-cell, urban micro-cell street canyon and urban macro-cell).

As future work, it could be useful to investigate the effect of the DS with different channel models and scenarios for 5G communication systems.

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